PYTHON ASSIGNMENT-1

ASSIGNMENT TITLE: Python Assignment 1: Data Structures - Strings & Tuples

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GITHUB LINK:

String Concatenation

Take string 1 = "Hello " and string 2 as the name entered by the user, concatenate them, then concatenate string 3 ", welcome to Python programming".

Program:

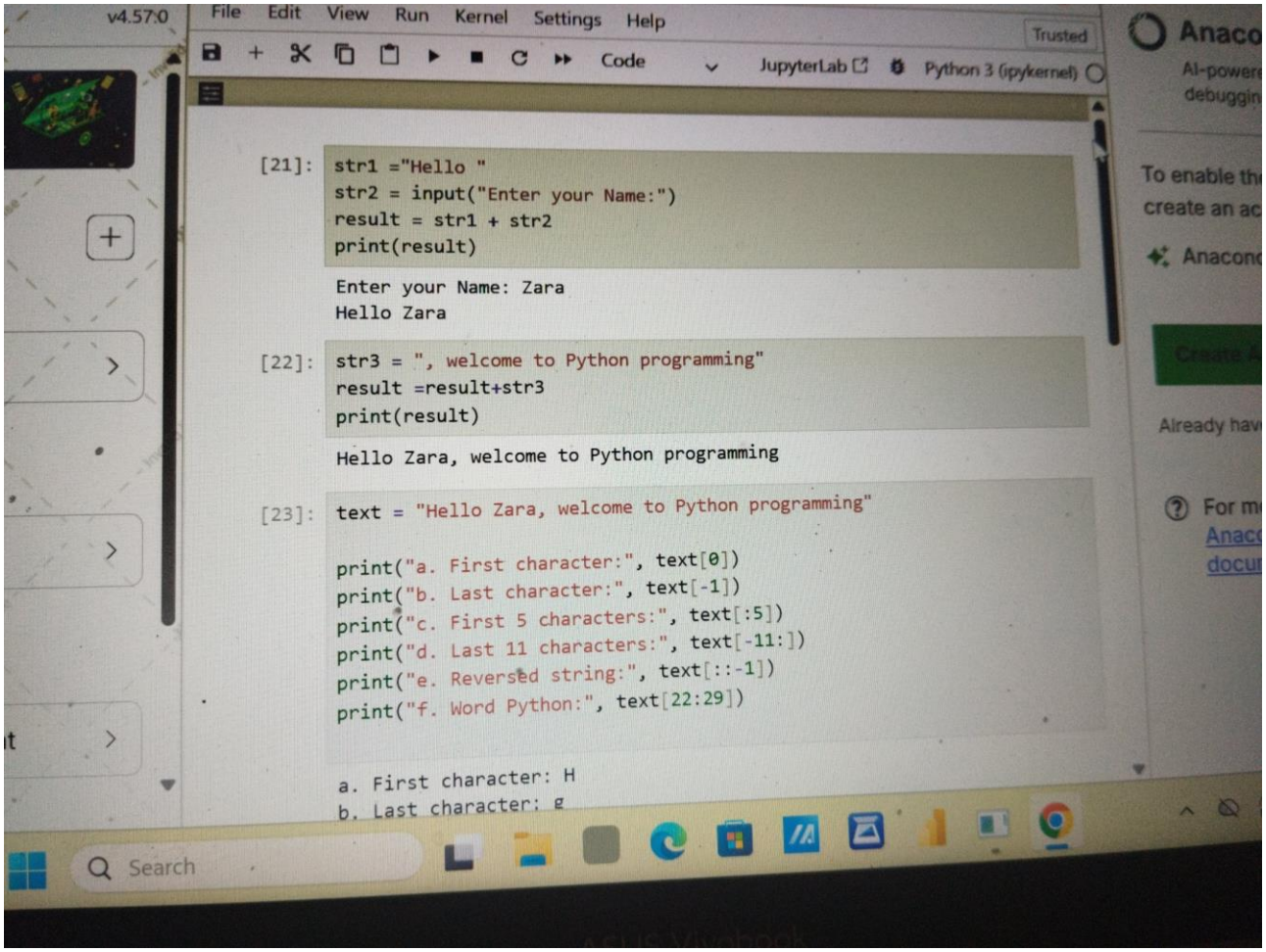
```
str1 = "Hello "
str2 = input("Enter your Name:")
result = str1 + str2
print(result)

str3 = ", welcome to Python programming"
result = result + str3
print(result)
```

Explanation:

- **input()** reads the name typed by the user as a string.
- The **+** operator joins (concatenates) two strings. The space after "Hello" keeps a gap before the name.
- **result = result + str3** adds string 3 to the existing string and stores it back in result.

Output:


v4.57.0
File Edit View Run Kernel Settings Help
Trusted
Code
JupyterLab
Python 3 (ipykernel)
[21]:
str1 ="Hello "
str2 = input("Enter your Name:")
result = str1 + str2
print(result)
Enter your Name: Zara
Hello Zara
[22]:
str3 = ", welcome to Python programming"
result =result+str3
print(result)
Hello Zara, welcome to Python programming
[23]:
text = "Hello Zara, welcome to Python programming"
print("a. First character:", text[0])
print("b. Last character:", text[-1])
print("c. First 5 characters:", text[:5])
print("d. Last 11 characters:", text[-11:])
print("e. Reversed string:", text[::-1])
print("f. Word Python:", text[22:29])
a. First character: H
b. Last character: g
Search

1. String Slicing and Indexing

Using the concatenated string, print the required characters and slices.

Program:

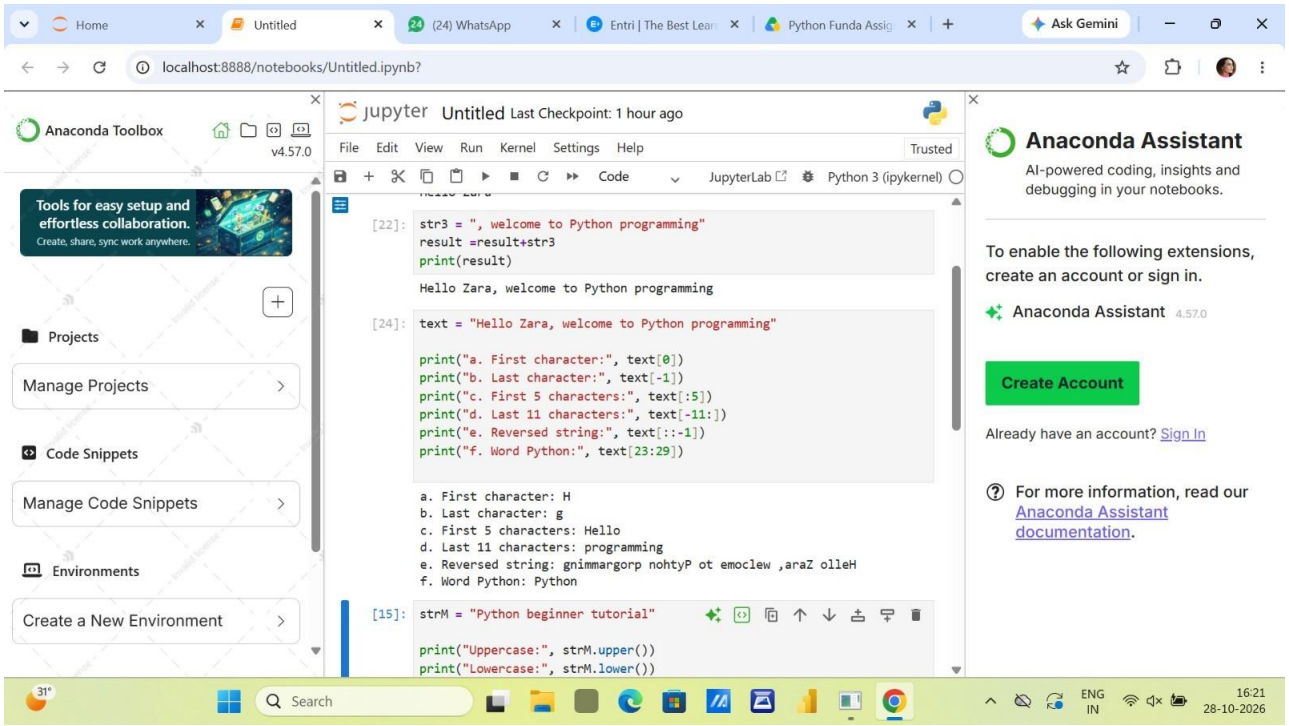
```
text = "Hello Zara, welcome to Python programming"

print("a. First character:", text[0])
print("b. Last character:", text[-1])
print("c. First 5 characters:", text[:5])
print("d. Last 11 characters:", text[-11:])
print("e. Reversed string:", text[::-1])
print("f. Word Python:", text[23:29])
```

Explanation:

- Indexing starts at 0, so **text[0]** is the first character; negative index **text[-1]** counts from the end.
- **text[:5]** gives the first 5 characters; **text[-11:]** gives the last 11 characters ("programming").
- **text[::-1]** uses step -1, which reverses the string.
- "Python" starts at index 23 and has 6 letters, so **text[23:29]** extracts it (end index 29 is excluded).

Output:



2. String Methods

Perform the string method tasks on strM = "Python beginner tutorial".

Program:

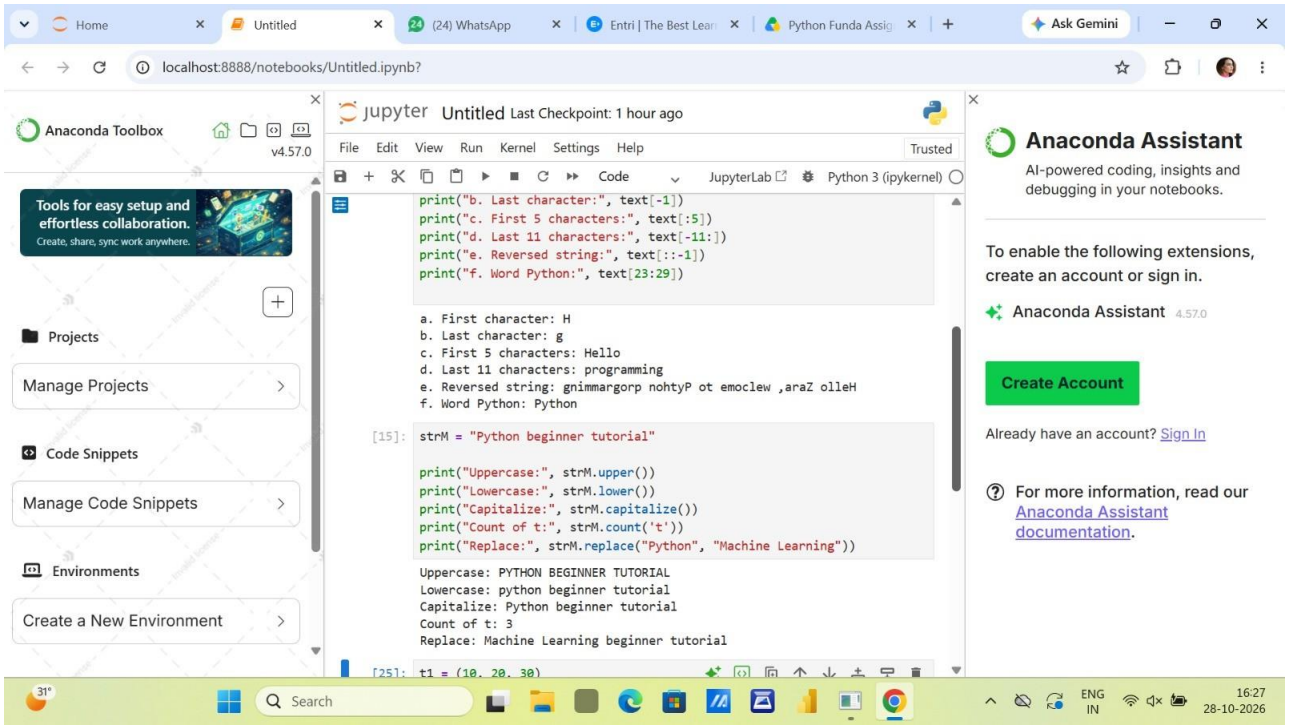
```
strM = "Python beginner tutorial"

print("Uppercase:", strM.upper())
print("Lowercase:", strM.lower())
print("Capitalize:", strM.capitalize())
print("Count of t:", strM.count('t'))
print("Replace:", strM.replace("Python", "Machine Learning"))
```

Explanation:

- **upper()** converts all letters to uppercase and **lower()** converts all letters to lowercase.
- **capitalize()** makes the first letter capital and the rest small, which gives the original form of the sentence.
- **count('t')** counts the lowercase letter t: once in "Python" and twice in "tutorial", so the answer is 3.
- **replace(old, new)** replaces every "Python" with "Machine Learning".

Output:



Tuples (Creation, Modification and Access)

Create t1 = (10, 20, 30) and t2 = (40, 50, 60) and perform the tasks a to e.

Program:

```
t1 = (10, 20, 30)
t2 = (40, 50, 60)

t_combine = t1 + t2
print("Combined tuple:", t_combine)
print("Repeated tuple:", t_combine * 3)
print("3rd element:", t_combine[2])
print("First three elements:", t_combine[:3])
print("Last three elements:", t_combine[-3:])
```

Explanation:

- Tuples are immutable, so **t1 + t2** creates a new tuple t_combine instead of changing the old ones.
- **t_combine * 3** repeats all the elements 3 times.
- The 3rd element is at index 2, so **t_combine[2]** = 30.
- **[:3]** gives the first three elements and **[-3:]** gives the last three elements.

Output:

